AI SYSTEMS PORTFOLIO · MARCH 2026

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Autonomous Agent Architecture & AI Systems Design

PROJECT DUCA

DucaBrain · Ubuntu 24.04
OpenClaw 2026.3.13 · Ollama
Johnstown, PA · March 2026

01 Project Overview

Project Duca is a fully autonomous, locally-hosted AI executive job search pipeline designed, built, debugged, and operationalized by AJ Sottile. Running on a purpose-built Ubuntu 24.04 workstation, the system deploys a six-agent AI pipeline powered by locally-running large language models via Ollama — with zero cloud dependency and zero paid API usage.

The system autonomously scouts job leads, vets companies, analyzes fit, writes tailored application materials, audits quality, and delivers a structured Morning Report every day at 8:00 AM ET. All outputs are filed to Google Drive. Duca communicates exclusively via Discord.

MISSION TARGET

Enterprise Enablement & Systems Architecture

Targeting strategic leadership roles at the intersection of organizational development and AI integration. Remote or Hybrid operational models with ≤25% travel.

DEPLOYMENT MODEL

100% Local · Zero Cloud

All models run locally on DucaBrain. No paid APIs. No cloud inference. Data never leaves the machine except scheduled Google Drive outputs shared with the operator.

COMMUNICATION

Discord Bot Channel

Duca operates as a Discord bot in a private guild. All operator communication, approvals, and reporting flow through a dedicated #general channel.

SCHEDULING

Daily at 8:00 AM ET

Morning Report fires automatically via OpenClaw's built-in Gateway scheduler. No external cron daemon required. Fully self-managing and persistent across reboots.

02 Six-Agent Pipeline

AGENT	ROLE	MODEL	RESPONSIBILITIES
Duca	Main Manager & Voice	Duca:latest	Operator interface, pipeline orchestration, Morning Report delivery, Discord
Agent 1 · Scout	Sub Coordinator	qwen3.5:4b	Job lead discovery across free sources, salary verification, Source of Truth URL confirmation
Agent 2 · Gatekeeper	Sub Researcher	qwen3.5:4b	URL verification, ghost job detection, fraud screening, Blacklist Log maintenance
Agent 3 · Analyst	Sub Analyst	qwen3.5:9b	Semantic JD analysis, pain point extraction, Match Score calculation (≥95% threshold)
Agent 4 · Copywriter	Sub Writer	qwen3.5:9b	Tailored cover letters (≤350 words), resume variant production, application packet assembly
Agent 5 · Inspector	Sub Auditor	qwen3.5:4b	Full QA audit, hallucination check, Drive packet creation, SQLite Vault logging

03 Technology Stack

AGENT FRAMEWORK OpenClaw 2026.3.13	LOCAL INFERENCE Ollama	PRIMARY MODEL Duca:latest (custom)	ANALYST / WRITER qwen3.5:9b
SUB-AGENTS qwen3.5:4b	EMBEDDINGS nomic-embed-text	COMMUNICATION Discord Bot API	OUTPUT STORAGE Google Drive (gog CLI)
HOST OS Ubuntu 24.04	SERVICE MANAGER systemd	SCHEDULER OpenClaw Cron Gateway	GOOGLE AUTH OAuth 2.0 / gogcli v0.12

04 Critical Debugging Achievement

ROOT CAUSE ANALYSIS — SILENT SYSTEM FAILURE

The Problem: When `api: "ollama"` is set in OpenClaw configuration, the framework sends the system prompt using `role: "developer"`. Ollama silently drops any message with an unrecognized role — no error, no warning, no log entry. Duca's 24,000-character system prompt vanished on every single request. Tool calls worked because they use `role: "tool"` — which is why file reads succeeded but full agent reasoning never followed.

The Fix: Switching to `api: "openai-completions"` with `baseUrl: "http://127.0.0.1:11434/v1"` and adding an explicit `compat` block — `supportsDeveloperRole: false, supportsReasoningEffort: false` — to the Duca model entry in both config files. This routes through OpenClaw's compatibility layer which correctly sends the system prompt as `role: "system"`.

Verification: Discord test "Duca, who are you?" returned full identity, mission awareness, 5-agent pipeline knowledge, Drive output structure, blacklist protocol, and approval requirements — complete system prompt reception confirmed on first attempt after fix.

05 Key Accomplishments

AI-ASSISTED DEVELOPMENT APPROACH

- Leveraged Google Gemini as the primary AI collaborator throughout initial system design, agent architecture planning, workspace file authoring, and early build phases
- Engaged Anthropic Claude for advanced diagnostic work, root cause analysis, configuration surgery, Google Workspace OAuth integration, and final operational hardening
- Demonstrated practical multi-model AI workflow — selecting the right AI tool for each phase of a complex technical project and directing both effectively toward a working production system

SYSTEM DESIGN & ARCHITECTURE

- Designed and deployed a purpose-built AI workstation (DucaBrain) running Ubuntu 24.04, configured exclusively for autonomous agentic AI workloads
- Architected a six-agent pipeline with role-separated responsibilities, model tier assignments, tool access policies, and isolated workspace and session stores per agent
- Implemented a custom Duca model (Duca:latest) built on qwen3.5:9b with a 32,768-token context window tuned for executive job search reasoning
- Designed a structured five-folder Google Drive output system with per-job packet generation and automatic operator sharing

CONFIGURATION & INTEGRATION

- Configured OpenClaw 2026.3.13 multi-agent system with per-agent model assignments, tool deny lists, workspace paths, and agent directory structures
- Integrated Discord as the primary communication channel — guild routing, channel allowlisting, bot identity, and message acknowledgment
- Authenticated Google Workspace via OAuth 2.0 using gogcli v0.12 on a headless Linux server using manual callback flow
- Configured file-based keyring and persisted GOG_KEYRING_PASSWORD to ~/.openclaw/.env for unattended gateway service execution
- Scheduled daily Morning Report cron job at 8:00 AM ET using OpenClaw's built-in Gateway scheduler with isolated session execution and Discord delivery

DEBUGGING & ROOT CAUSE ANALYSIS

- Identified a silent API adapter incompatibility between OpenClaw's ollama mode and Ollama's message role handling — a failure mode with no error output of any kind
- Cross-referenced source code and a published Medium post documenting the identical issue to confirm root cause before touching any files
- Applied targeted three-field fix to two separate config files with full backup protocol, post-edit verification, and gateway restart — zero data loss, first-attempt success
- Maintained strict research-before-action protocol throughout: no guesses, no assumptions, help files and official documentation consulted before every command

06 Technical Competencies Demonstrated

AI / LLM SYSTEMS

- Local LLM deployment and management (Ollama)
- Multi-agent system architecture and orchestration
- System prompt design and context engineering
- Agent role separation and tool policy design
- Embedding-based memory systems
- Model selection and parameter tuning
- API adapter compatibility analysis

INFRASTRUCTURE & INTEGRATION

- Linux server administration (Ubuntu 24.04)
- systemd service configuration and management
- JSON configuration management at scale
- OAuth 2.0 headless authentication flows
- Google Workspace API integration
- Discord bot configuration and channel routing
- Environment variable security and secret management

PROBLEM SOLVING

- Silent failure diagnosis without log output
- Source code analysis for root cause identification
- Cross-referencing documentation and community sources
- Systematic backup-verify-restore protocols
- Research-first, protocol-driven methodology

AI TOOLS & FRAMEWORKS

- OpenClaw 2026 — multi-agent gateway framework
- Ollama — local model serving and management
- gogcli v0.12 — Google Workspace CLI automation
- Qwen 3.5 model family (4b, 9b variants)
- Gemini (Google) — design and build phases
- Claude (Anthropic) — debugging and integration
- Discord API — bot deployment and management

07 Operational Status

COMPLETED

Core System Fix — API Adapter Correction

Both config files updated and verified. Duca confirmed receiving full system prompt. Discord test confirmed complete mission awareness on first attempt.

COMPLETED

Google Workspace Authentication

OAuth 2.0 headless flow completed for Duca.AJS@gmail.com. File-based keyring configured. All Google services authenticated and operational.

COMPLETED

Google Drive — Source Documents & Operator Access

Resume (PDF + DOCX), Cover Letter template, Contact Info, and Blacklist Log all confirmed visible to Duca. All 5 output folders and root shared with operator at editor level.

COMPLETED

Morning Report Scheduling

Cron job active in OpenClaw Gateway scheduler: daily at 8:00 AM America/New_York. GOG_KEYRING_PASSWORD persisted for unattended execution.

LIVE

System Fully Operational

Gateway running, Discord connected, all six agents loaded, cron armed. First Morning Report fires tomorrow at 8:00 AM ET.

- Gateway Running
- Discord Connected
- Google Drive Authenticated
- Cron Armed · 8:00 AM ET
- Docker Sandbox Ready
- All 6 Agents Loaded